

SAT Math Mastery · Week TP1M**SAT Trick Pack 1 — Part M****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** finite differences · pyramid slant height · exponential forms · vertex-time ratios
· SSA triangles · symmetric vertex form · identity parameters · circle tangency
· tangent lengths · discriminant parameters · percent chains · exponent match-
ing · mixtures · rational expressions**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Extend a quadratic table with first and second differences instead of solving for coefficients.
- S2.** Chain base area $\rightarrow$ side $\rightarrow$ height $\rightarrow$ slant height $\rightarrow$ face area in a pyramid.
- S3.** Rewrite a shifted exponential so the starting value is an explicit coefficient.
- S4.** Compare two drops from a maximum to locate the vertex of a parabola.
- S5.** Count triangles in SSA data by comparing the opposite side with $b \sin A$ and with b .
- S6.** Use equal outputs to fix the axis of symmetry, then vertex form.
- S7.** Make a linear equation an identity: match x -coefficients and constants.
- S8.** Tangency by distance from the centre to the line; keep both signs of the absolute value.
- S9.** Chord, central angle and tangent length from the special triangle at the centre.
- S10.** Zero discriminant gives a relation between parameters; test every sign case against a side condition.
- S11.** Compose percent changes as multipliers from a convenient base of 100.
- S12.** Put both variables over one base and equate exponents.
- S13.** Translate a relational phrase into one expression and use the conserved total.
- S14.** Simplify a complex fraction with difference of squares, then use the given sum and product.

2 · REQUIRED FACTS

- For equally spaced inputs, a quadratic has constant second differences, equal to $2ah^2$ for step h .
- For a regular pyramid with a square base, the slant height satisfies $\ell^2 = h^2 + \left(\frac{s}{2}\right)^2$; $V = \frac{1}{3}Bh$.
- In SSA with acute angle A : no triangle if $a < b \sin A$; one (right) triangle if $a = b \sin A$; two if $b \sin A < a < b$; one if $a \geq b$.
- A circle tangent to both axes has centre $(\pm r, \pm r)$; its general-form constant is $F = r^2$ when the centre is on the line $y = \pm x$.
- Tangent segments from one external point are equal; a tangent is perpendicular to the radius at the point of contact.
- A quadratic has exactly one real zero when its discriminant is 0.
- $p\%$ more means multiply by $1 + \frac{p}{100}$; $p\%$ less means multiply by $1 - \frac{p}{100}$.
- The distance from (x_0, y_0) to $Ax + By + C = 0$ is $\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$.

3 · REQUIRED FORMULAS

$$\Delta_2 = 2ah^2, \quad \ell = \sqrt{h^2 + \left(\frac{s}{2}\right)^2}, \quad h_{\text{SSA}} = b \sin A, \quad f(x) = a(x - h)^2 + k, \quad h = \frac{p+q}{2} \text{ if } f(p) = f(q)$$

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}, \quad b^2 - 4ac = 0, \quad \frac{1}{x} - \frac{1}{y} = \frac{y-x}{xy}, \quad u^2 - v^2 = (u-v)(u+v), \quad PA = r \tan \frac{\theta}{2}$$

4 · MINI THEORY SHEET

Tables. Equally spaced inputs: subtract to get first differences, subtract again to get a constant second difference, then extend the pattern one row at a time. No coefficients are needed.

Parabolas. Equal outputs at two inputs put the axis at their midpoint. Two drops below the maximum are in the ratio of the squares of the distances to the vertex time, so the unknown scale constant cancels.

Parameters. Identity: both coefficients and constants match. Tangency: distance = radius gives $|\dots| = \text{number}$, which has two cases. Zero discriminant: take both square-root signs, then use the side condition to discard one.

Chains. Start every percent chain from 100 and multiply along it; ask which quantity is the base of the final comparison.

Exponents. Write each variable as a power of one base before combining; compare exponents only at the end.

Triangles. Draw the height $b \sin A$ from the vertex; the opposite side either misses it, touches it, or swings to two positions.

Speed shortcuts.

Situation	Shortcut
equally spaced table of a quadratic	first, second differences; extend; no system of equations
pyramid with base area and volume	$s = \sqrt{B}$, $h = \frac{3V}{B}$, slant from the right triangle with legs h and $\frac{s}{2}$
decaying exponential, $t \geq 0$	choose the form whose exponent is 0 at $t = 0$; the coefficient is the maximum
two heights below a known maximum	ratio of drops = ratio of squared distances; solve for the vertex time
SSA data	compare a with $b \sin A$ and with b ; count solutions
$f(p) = f(q)$	vertex at $x = \frac{p+q}{2}$, use $a(x-h)^2 + k$
“for all x ” with parameters	expand, match coefficients and constants
circle and line touch once	distance from centre to line = r ; both signs
chord at a given distance, central angle θ	isosceles triangle at the centre; tangent length = $r \tan \frac{\theta}{2}$
one real solution, two parameters	$b^2 = 4ac$; test both square-root signs
chain of percents	assign 100, follow multipliers
$a^m = b^n$	$a = b^{n/m}$, then substitute
mixture with a relational clue	one variable, one total
$\frac{1}{x} - \frac{1}{y}$ over $\frac{1}{x^2} - \frac{1}{y^2}$	factor the difference of squares and cancel, then use $x + y$ and xy

5 · GUIDED EXAMPLES

Example 1 — Extend a table with differences

A quadratic function g has $g(0) = 5$, $g(2) = 11$, $g(4) = 23$, $g(6) = 41$. Find $g(10)$.

First differences: 6, 12, 18. Second differences: 6, 6 (constant). Next first differences: 24, then 30.
 $g(8) = 41 + 24 = 65$ and $g(10) = 65 + 30 = 95$.

Example 2 — Tangency gives two cases

The line $3x - 4y = k$ is tangent to $x^2 + y^2 - 4x + 6y = 12$. Find the greatest possible k .

Centre $(2, -3)$, $r^2 = 12 + 4 + 9 = 25$, so $r = 5$. Distance: $\frac{|3(2) - 4(-3) - k|}{5} = 5 \Rightarrow |18 - k| = 25$.
 $k = 43$ or $k = -7$. The greatest is 43.

Example 3 — Percent chain from a base of 100

a is 20% greater than b , and b is 25% less than c . What percent of c is a ?

Let $c = 100$: $b = 75$, $a = 1.2(75) = 90$. So a is 90% of c .

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f satisfies $f(-1) = f(7)$, $f(0) = -4$ and $f(4) = 12$. What is the value of $f(10)$?

(A) -98

(B) -84 ✓

(C) 0

(D) 98

1. Equal outputs at $x = -1$ and $x = 7$ put the axis of symmetry at $x = \frac{-1+7}{2} = 3$. Write $f(x) = a(x-3)^2 + k$.

2. $f(0) = -4$: $9a + k = -4$. $f(4) = 12$: $a + k = 12$.

3. Subtract: $8a = -16$, so $a = -2$, and then $k = 14$. So $f(x) = -2(x-3)^2 + 14$.

4. $f(10) = -2(7)^2 + 14 = -98 + 14 = -84$.

5. (Options -98 drops k ; 0 forgets to square.)

Answer: (B)

2. SAT CHALLENGE · NO CALCULATOR NEEDED

Quantity Q is 60% greater than quantity P . Quantity R is 25% less than Q . Quantity R is 50% greater than quantity S . P is what percent greater than S ?

1. Let $P = 100$. Then $Q = 1.6(100) = 160$ and $R = 0.75(160) = 120$.

2. $R = 1.5S$, so $S = \frac{120}{1.5} = 80$.

3. P compared with S : $\frac{100-80}{80} = 0.25$, i.e. 25% greater.

4. (The base of the comparison is S ; 20% would be the percent by which S is less than P .)

Answer: 25

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

For nonzero real numbers p and q , the equation $px^2 + 3qx + 4p = 0$ has exactly one real solution, and $pq = -48$. If $p > 0$, what is the value of $p + q$?

(A) 14 (B) -14

(C) -2 ✓ (D) 2

1. Exactly one real solution means the discriminant is 0: $(3q)^2 - 4(p)(4p) = 0$, so $9q^2 = 16p^2$.

2. Taking square roots: $3q = 4p$ or $3q = -4p$, i.e. $q = \frac{4p}{3}$ or $q = -\frac{4p}{3}$.

3. If $q = \frac{4p}{3}$ then $pq = \frac{4p^2}{3} > 0$, contradicting $pq = -48$. So $q = -\frac{4p}{3}$.

4. Then $pq = -\frac{4p^2}{3} = -48 \Rightarrow p^2 = 36$. Since $p > 0$, $p = 6$ and $q = -8$.

5. $p + q = 6 - 8 = -2$. (Option 14 comes from the rejected case $q = 8$.)

Answer: (C)

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle PQR , the measure of angle P is 30° and $PR = 14$. The length $QR = a$ is a positive integer with $a \leq 20$. For how many values of a does the given information determine exactly one triangle, up to congruence?

1. Here the known angle is $P = 30^\circ$, the adjacent side is $PR = 14$ and the opposite side is $QR = a$ (an SSA situation). The height is $14 \sin 30^\circ = 7$.

2. $a < 7$: side QR is too short to reach line PQ , no triangle. $a = 7$: it just touches, one right triangle.

3. $7 < a < 14$: two different triangles (two positions for Q). $a \geq 14$: only one position, since the second intersection lies on the opposite side of P .

4. Exactly one triangle: $a = 7$ and $a = 14, 15, \dots, 20$, which is $1 + 7 = 8$ values.

Answer: 8

5.

SAT CHALLENGE · DESMOS

In the xy -plane, the circle $x^2 + y^2 + 8x - 2y = 8$ and the line $5x - 12y = k$, where k is a constant, intersect at exactly one point. What is the greatest possible value of k ?

(A) 33 ✓

(B) -97

(C) 65

(D) 32

1. Complete the square: $(x + 4)^2 + (y - 1)^2 = 8 + 16 + 1 = 25$. Centre $(-4, 1)$, radius 5.

2. One intersection point means the line is tangent, so the distance from the centre to the line equals 5: $\frac{|5(-4) - 12(1) - k|}{\sqrt{5^2 + 12^2}} = 5$.

3. $\frac{|-32 - k|}{13} = 5 \Rightarrow |k + 32| = 65$.

4. $k + 32 = 65$ gives $k = 33$; $k + 32 = -65$ gives $k = -97$. The greatest is 33.

5. (Option -97 is the other case; 65 and 32 come from not finishing the absolute-value equation.)

Answer: (A)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

The table gives values of a quadratic function g at equally spaced inputs.

x	2	5	8	11
$g(x)$	44	95	200	359

What is the value of $g(17)$?

(A) 677

(B) 572

(C) 785

(D) 839 ✓

1. Inputs step by 3. First differences of g : $95 - 44 = 51$, $200 - 95 = 105$, $359 - 200 = 159$.

2. Second differences: $105 - 51 = 54$ and $159 - 105 = 54$, constant, as expected for a quadratic.

3. Next first difference: $159 + 54 = 213$, so $g(14) = 359 + 213 = 572$.

4. The one after that: $213 + 54 = 267$, so $g(17) = 572 + 267 = 839$.

5. (Check: $g(x) = 3x^2 - 4x + 40$ gives $g(17) = 867 - 68 + 40 = 839$. Option 572 is $g(14)$; 677 keeps the last difference 159 constant; 785 adds 213 twice.)

Answer: (D)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$k(3x - 2) - (x - m) = 8x + 10$$

In the equation above, k and m are constants. The equation is true for every real value of x . What is the value of $k + m$?

1. Expand the left side: $3kx - 2k - x + m = (3k - 1)x + (m - 2k)$.
2. True for every x means the x -coefficients match and the constants match: $3k - 1 = 8$ and $m - 2k = 10$.
3. $k = 3$, then $m = 10 + 2(3) = 16$.
4. $k + m = 3 + 16 = 19$.

Answer: 19

8.

SAT CHALLENGE · CALCULATOR

The concentration of a medication in the bloodstream, in milligrams per liter, is modeled by $C(t) = 18 \left(\frac{1}{3}\right)^{\frac{t-4}{2}}$, where $t \geq 0$ is the number of hours after the first measurement. Which of the following is an equivalent form of $C(t)$ that displays the maximum value of C as a constant or coefficient?

- (A) $486 \cdot 3^{-\frac{t+2}{2}}$ (B) $18 \cdot 3^{\frac{4-t}{2}}$
 (C) $162 \cdot 3^{-\frac{t}{2}}$ ✓ (D) $54 \cdot 3^{\frac{2-t}{2}}$

1. $\left(\frac{1}{3}\right)^{\frac{t-4}{2}} = 3^{\frac{4-t}{2}}$, so C is a decreasing function of t and its maximum on $t \geq 0$ occurs at $t = 0$.
2. $C(0) = 18 \cdot 3^2 = 162$.
3. Split the exponent: $18 \cdot 3^{\frac{4-t}{2}} = 18 \cdot 3^2 \cdot 3^{-\frac{t}{2}} = 162 \cdot 3^{-\frac{t}{2}}$. The coefficient is the value at $t = 0$, i.e. the maximum.
4. The other options are equivalent too, but their exponents equal 2, 1 and -1 at $t = 0$, so their coefficients (18, 54, 486) are not the maximum.

Answer: (C)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

In a circle with center O , chord $\overline{AB}$ has $\angle AOB = 120^\circ$ and lies at a perpendicular distance of 3 from O . The lines tangent to the circle at A and at B intersect at point P . What is the length of $\overline{PA}$?

(A) 6

(B) $6\sqrt{3}$ ✓

(C) 12

(D) $3\sqrt{3}$

1. Triangle OAB is isosceles with apex angle 120° . The perpendicular from O to AB bisects the apex angle, giving a right triangle with a 60° angle at O , so $3 = r \cos 60^\circ$ and $r = 6$.
2. A tangent is perpendicular to the radius, so $\angle OAP = 90^\circ$. By symmetry OP bisects $\angle AOB$, so $\angle AOP = 60^\circ$.
3. $PA = r \tan 60^\circ = 6\sqrt{3}$. (Equal tangent segments: $PB = PA$.)
4. (Option $3\sqrt{3}$ treats the distance 3 as the radius; 6 is r ; 12 is OP .)

Answer: (B)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and the volume of the pyramid is 400 cubic inches. What is the combined area, in square inches, of the four triangular lateral faces?

(A) 240

(B) 130

(C) 360

(D) 260 ✓

1. Base area 100 gives side $s = 10$. From $V = \frac{1}{3}Bh$: $400 = \frac{1}{3}(100)h$, so the vertical height is $h = 12$.
2. The height of a triangular face (the slant height) is the hypotenuse of a right triangle with legs $h = 12$ and $\frac{s}{2} = 5$:
 $\ell = \sqrt{144 + 25} = 13$.
3. Each face has area $\frac{1}{2}(10)(13) = 65$. Four faces: $4 \cdot 65 = 260$.
4. (Option 240 uses the vertical height 12 as the face height; 130 is two faces; 360 adds the base.)

Answer: (D)

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

A 84-gram sample of an alloy contains only copper, zinc and tin. The mass of the copper is 5 grams more than twice the mass of the zinc. The mass of the tin is half the combined mass of the copper and the zinc. What is the mass, in grams, of the tin in the sample?

1. Let z be the mass of zinc. Copper: $2z + 5$. Combined copper and zinc: $3z + 5$, so tin is $\frac{3z + 5}{2}$.

2. Total: $(3z + 5) + \frac{3z + 5}{2} = \frac{3}{2}(3z + 5) = 84 \Rightarrow 3z + 5 = 56$.

3. Tin = $\frac{56}{2} = 28$ grams. (Check: $z = 17$, copper 39, tin 28, total 84.)

Answer: 28

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers a and b , both greater than 1, satisfy $a^3 = b^2$. If $\frac{a^9}{\sqrt[3]{b}} = b^k$, what is the value of k ?

(A) $\frac{17}{3}$ ✓

(B) 6

(C) $\frac{19}{3}$

(D) $\frac{17}{2}$

1. From $a^3 = b^2$: $a^9 = (a^3)^3 = (b^2)^3 = b^6$.

2. $\sqrt[3]{b} = b^{1/3}$, so $\frac{a^9}{\sqrt[3]{b}} = \frac{b^6}{b^{1/3}} = b^{6 - \frac{1}{3}} = b^{17/3}$.

3. Since $b > 1$, equal powers of b have equal exponents: $k = \frac{17}{3}$.

4. (Option 6 ignores the cube root; $\frac{19}{3}$ adds instead of subtracting $\frac{1}{3}$.)

Answer: (A)

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

A water balloon is launched upward from a rooftop. Its height above the ground, in feet, t seconds after launch is modeled by a quadratic function h . The maximum height of 125 feet is reached at a time between $t = 1$ and $t = 6$, and $h(1) = 80$ and $h(6) = 105$. What is the height, in feet, of the balloon at the moment of launch?

(A) 80

(B) 45 ✓

(C) 75

(D) 37

- Write $h(t) = 125 - a(t - v)^2$, where v is the time of maximum height. The drops below the maximum are $125 - 80 = 45$ at $t = 1$ and $125 - 105 = 20$ at $t = 6$.
- Divide: $\frac{a(v - 1)^2}{a(6 - v)^2} = \frac{45}{20} = \frac{9}{4}$, so $\frac{v - 1}{6 - v} = \frac{3}{2}$ (the maximum lies between the two times).
- $2v - 2 = 18 - 3v \Rightarrow v = 4$. Then $45 = a(3)^2 \Rightarrow a = 5$.
- $h(0) = 125 - 5(4)^2 = 125 - 80 = 45$.
- (Option 75 continues the average rate of change from $t = 1$ to $t = 6$ backwards; 80 is $h(1)$; 37 places the vertex halfway at $t = 3.5$.)

Answer: (B)**14.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a circle is tangent to both the x -axis and the y -axis, has its center in the second quadrant, and passes through the point $(-2, 9)$. The diameter of the circle is less than 15. The circle has equation $x^2 + y^2 + Dx + Ey + F = 0$, where D , E and F are constants. What is the value of F ?

- Tangent to both axes with center in the second quadrant: center $(-r, r)$, radius $r > 0$. Equation: $(x + r)^2 + (y - r)^2 = r^2$.
- Expand: $x^2 + y^2 + 2rx - 2ry + r^2 = 0$, so $F = r^2$.
- Point $(-2, 9)$: $(-2 + r)^2 + (9 - r)^2 = r^2 \Rightarrow r^2 - 22r + 85 = 0 \Rightarrow (r - 5)(r - 17) = 0$.
- Diameter < 15 means $r < 7.5$, so $r = 5$ and $F = 25$. (The root $r = 17$ gives diameter 34.)

Answer: 25

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x^2} - \frac{1}{y^2}} ?$$

(A) 1

(B) 4

(C) $\frac{12}{5}$ ✓(D) $\frac{5}{12}$

1. Factor the denominator as a difference of squares: $\frac{1}{x^2} - \frac{1}{y^2} = \left(\frac{1}{x} - \frac{1}{y}\right) \left(\frac{1}{x} + \frac{1}{y}\right)$.

2. Since $x \neq y$, $\frac{1}{x} - \frac{1}{y} \neq 0$ and cancels, leaving $\frac{1}{\frac{1}{x} + \frac{1}{y}} = \frac{1}{\frac{x+y}{xy}} = \frac{xy}{x+y}$.

3. $\frac{xy}{x+y} = \frac{36}{15} = \frac{12}{5}$. (Check with $x = 12$, $y = 3$: $\frac{\frac{1}{12} - \frac{1}{3}}{\frac{1}{144} - \frac{1}{9}} = \frac{-\frac{1}{4}}{-\frac{5}{48}} = \frac{12}{5}$.)

4. (Option $\frac{5}{12}$ is the reciprocal.)

Answer: (C)